

Dive into AKLab

Andreas Kempa-Liehr

Department of Engineering Science and Biomedical Engineering
The University of Auckland

Welcome to AKLab!

1 Consider the big picture

In the next weeks and months, we are going to work on some exciting research projects and the ultimate goal is to produce cutting edge research results, which have the potential to be published.

The journey starts now and you should invest 30 minutes to brainstorm, how you are going to

- make notes,
- collect and store data,
- manage your source code,
- communicate insights and problems to your collaborators.

Remember, some notes, which you are collecting in the very beginning of the project, might come handy, when you are writing everything up. So, think about your notes as *messages to your future-self*.

2 Project meetings

You are going to lead the weekly project meetings. The goal is to use your collaborator's time as efficiently as possible. This exercise simulates a typical professional situation, where an analyst reports progress, challenges, questions, and feedback to a team lead or customer.

The day before the meeting, you should email the meeting agenda to your collaborators. The meeting agenda is a bullet point list with topics and questions to be covered in the upcoming meeting. It also confirms time and location of the meeting. After the meeting, you are expected to share the minute notes with your collaborators. The minute notes cover major discussion points and next steps for the upcoming week.

The agenda and minute notes are an important part of the your personal knowledge management and complement your personal weekly review of your project's progress. [Heylighen and Vidal \(2008\)](#) give a concise description how these reviews enable stress-free productivity.

3 Best practises

Work through the following best practises for reproducible research and reflect how these recommendations map to your approach. Are you missing something?

Wilson et al. (2014) Best Practices for Scientific Computing.

Wilson et al. (2017) Good enough practices in scientific computing.

Paulus and Biskup (2023) Towards more reproducible and FAIRer research data.

Kempa-Liehr (2024) The Python module `contexere`¹ implements a naming convention for research artefacts and is an update of the naming convention presented by Kühne and Liehr (2009).

Kühne and Liehr (2009) Improving the Traditional Information Management in Natural Sciences.

Riede et al. (2010) On the Communication of Scientific Data: The Full-Metadata Format.

Silyn-Roberts (2013) provides an excellent overview on writing scientific documents.

Hunt (2008) gives an engineers perspective on knowledge management and creativity.

4 Setup and manage your research environment

Setup your project [Home - Cookiecutter Data Science](#)

Make yourself familiar with Git [Git - Book](#)

Sourcetree is a nice Git GUI [Sourcetree | Free Git GUI for Mac and Windows](#)

Python coding From novice to competent programmer

Getting started Chapter 1 to 3 of [Moruzzi \(2020\)](#)

Improve your coding [gregmalcolm/python_koans: Python Koans - Learn Python through TDD](#)

5 Get started with Time Series Analytics

Learn about time-series features Read chapters 1 and 8 of [Nielsen \(2019\)](#) and the overview from [Fulcher \(2018\)](#)

Learn about tsfresh Read [Christ et al. \(2016, 2018\)](#)

Get started with tsfresh [blue-yonder/tsfresh: Automatic extraction of relevant features from time series:](#) by running the notebooks [tsfresh/notebooks](#) at master · [blue-yonder/tsfresh](#)

Review your coding happits [Coding habits for data scientists | ThoughtWorks](#)

¹[kempa-liehr/contextere](#): Naming convention for research artefacts

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